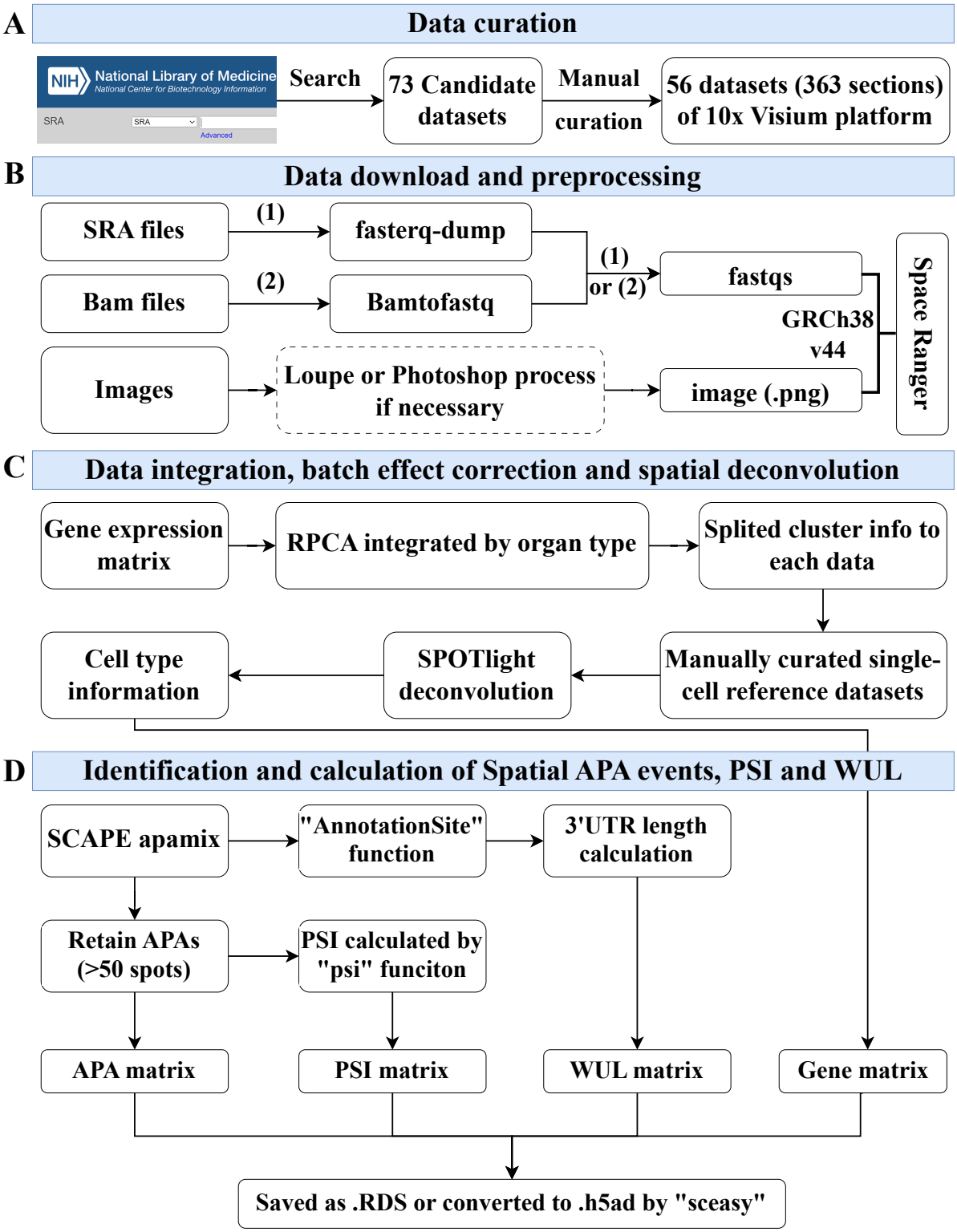
A
Data curation
National Library of Medicine
National Center for Biotechnology Information
SRA
Advanced
Search
73 Candidate datasets
Manual curation
56 datasets (363 sections) of 10x Visium platform
B
Data download and preprocessing
SRA files
(1)
fasterq-dump
Bam files
(2)
Bamtofastq
(1) or (2)
fastqs
GRCh38 v44
Images
Loupe or Photoshop process if necessary
image (.png)
Space Ranger
C
Data integration, batch effect correction and spatial deconvolution
Gene expression matrix
RPCA integrated by organ type
Splited cluster info to each data
Manually curated single-cell reference datasets
SPOTlight deconvolution
Cell type information
D
Identification and calculation of Spatial APA events, PSI and WUL
SCAPE apamix
"AnnotationSite" function
3'UTR length calculation
Retain APAs (>50 spots)
PSI calculated by "psi" funciton
APA matrix
PSI matrix
WUL matrix
Gene matrix
Saved as .RDS or converted to .h5ad by "sceasy"